

Active Scene Graphs: a scene graph that acts to keep itself true

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Abstract—A scene graph is normally a record: a data structure that perception fills and planning reads. We propose the *Active Scene Graph* (ASG), a scene graph that acts to keep itself true. Every node is a concept with a posterior and a precision; every concept publishes what the robot could do to reduce its own uncertainty; the robot’s body is one such concept and calibrates itself the same way; and a single controller ranks every offer in one currency, nats, and executes them in real time over a distributed graph. We state this as five requirements over standard scene graphs: actionable, epistemically self-constructing, concept-oriented, embedding a self-calibrating robot model, and real-time over a distributed architecture. Each requirement has prior art; the claim of this paper is that they are *coupled*, so a system that meets them must meet them together, and that we have one. We give the model (concept nodes with posterior, precision and existence; affordances as priced offers; one expected free energy with a task prior as the only difference between representing and acting), the architecture that meets the concurrency requirements (a CRDT graph, a zero-copy media plane, and two protocols verified in TLA+), and an evaluation of the parts that run, in simulation and on two physical robots. We report the results that support the requirements and the ones that do not: on 594 annotated floor plans the epistemic viewpoint selector is statistically indistinguishable from a coverage heuristic, and we explain why the endpoint could not have discriminated. A ledger separates what is designed, implemented and evaluated.

I. INTRODUCTION

Three-dimensional scene graphs have become the standard representation for robots that must reason about places, objects and their relations [1], [2], [3], [4]. In every one of these systems the graph is a *record*. Perception writes nodes and edges; planning reads them; the graph itself neither knows how uncertain it is nor does anything about it. When the record is wrong, nothing inside the representation notices, and when it is incomplete, completing it is somebody else’s job: an exploration module driven by a frontier, a task planner driven by a language model, or a person with a joystick.

This paper describes a scene graph that is not a record. In an *Active Scene Graph* the representation is the thing that acts. Every node is a *concept*: a belief with a mean, a covariance, a precision and a log-odds of existence, maintained by an agent that owns it. Every concept publishes, as nodes in the same graph, the actions the robot could take to reduce that concept’s uncertainty, priced in nats. The robot’s own body is a concept in the same sense, whose kinematic and extrinsic parameters are beliefs corrected online by the same machinery that corrects everything else. A controller ranks every published offer, whatever its source, under one expected free energy, and a task is nothing more than a prior preference

added to that functional. The representational layer and the task layer differ by one term.

We state the idea as five requirements (§II). Each has prior art, and we make no claim to be first on any of them. Our claim is different and, we argue, stronger: the requirements are *coupled*. An actionable offer needs a calibrated probability behind it; a calibrated probability needs a solved self-model; several writers need ownership; ownership needs a transport that guarantees cleanup; bounded bandwidth needs a graph that carries beliefs rather than pixels; and the self-model learns only because acting is the experiment. Satisfying the requirements one at a time does not compose. Every one of these dependencies was found the hard way, as a measured incident in a running system, and we report the incidents.

The contributions are: (i) the five requirements and the coupling argument (§II); (ii) the model: concept nodes with posterior, precision and existence, affordances as priced offers, and one expected free energy for both regimes (§IV, §V); (iii) the robot as a concept that calibrates itself from the corrections that ordinary actions generate (§VI); (iv) an architecture that meets the concurrency requirements, with two protocols verified in TLA+ (§VII); (v) an evaluation of the running parts in simulation and on two physical robots, including a null result and its cause (§VIII); and (vi) an honest ledger of what is designed, implemented and evaluated (Table IV).

This preprint extends our IWAI abstract [5], which stated the actionable-graph idea for an active-inference audience with a Fisher-information viewpoint selector, and the concept-first architecture of [6], which built rooms and doors as agents but with intentions rather than a formalisation: no stated joint, no priced offers, no lifecycle contract, no self-model. What is new here relative to both is the requirement set, the coupling argument, the single-currency pricing, the self-calibrating robot concept, and the evaluation.

II. FIVE REQUIREMENTS, AND WHY THEY ARE ONE

Table I states the requirements. The second column is the sentence the paper commits to; it is written so that it can be false. The third says what a Hydra- or ConceptGraphs-class system does instead. The fourth names the mechanism here. The last two columns give the status, using the ledger’s three letters (D designed, I implemented, E evaluated), and the evidence.

A. The coupling argument

A checklist can be satisfied one row at a time. The requirements above cannot, and the reason is a set of dependencies

TABLE I
THE FIVE REQUIREMENTS. D DESIGNED, I IMPLEMENTED, E EVALUATED.

R	Falsifiable statement	Standard scene graph	Mechanism here	Status	Evidence
R1 Actionable	Every node carries executable offers whose preconditions are probabilities read from the graph, and one generic executor runs any of them without knowing the concept.	Nodes are labels plus geometry; actions are planned outside the graph from a task specification.	Affordance nodes under a <code>has_intention</code> edge; epistemic offers (one per object) and pragmatic offers (named, preconditioned: approach, open, cross a door). Execution claim = epoch + an <code>executing</code> edge the consumer owns.	protocol I+E; door offers I; task-level EFE D	TLA+ model of the claim protocol; §VIII-C
R2 Epistemic self-construction	The graph's own posterior uncertainty is consumed: it gates publication and is priced into the offers the robot acts on. No hand thresholds in the belief pipeline.	Uncertainty, if tracked, is stored; exploration is a frontier heuristic.	Per-concept generative model with an adequacy target Σ^* ; next view from ΔH weighted by $P(\text{detect})$; existence as log-odds with absence weighted by $P(\text{detect})$ and occlusion holding; a structural adoption judge for the room.	I+E (objects, room); existence I	Calibrated corner σ on a physical robot; 50 synthetic rooms; 594 real layouts (null, §VIII-B)
R3 Concept-oriented	The graph is populated by independent concept agents, each owning one class of node under a normative lifecycle, and the fleet composes without a central planner.	One monolithic mapper; classes are labels on one pipeline.	Lifecycle contract (create, update, remove, history, ownership) with a per-agent conformance audit; ten concept agents plus meta-concepts and a long-term memory; an <code>owns</code> attribute ties node identity to agent identity.	I+E	Conformance audit; phantom births 130→1 (§VIII-D)
R4 Self-calibrating robot model	The robot is a node like any other: its kinematic and extrinsic parameters are beliefs updated online from the localiser's own correction, and their precision propagates into every offer's price.	The robot is a fixed URDF; calibration is offline.	Robot frame tree as a concept; batch estimator over episodes; closed-loop camera extrinsics; wheel-odometry covariance injection; motion preintegration.	I+E	Extrinsic yaw recovery 0.984; boresight +0.814° applied (§VIII-E)
R5 Real-time, distributed	N agents on M machines share one graph with bounded latency; sensor streams bypass the graph; agents join and leave without corrupting it.	A single process, or ROS topics with one map owner.	CRDT graph; zero-copy media plane on a dedicated DDS domain; debounced presence protocol; ownership cleanup on graceful exit.	I+E	Stream lag 35→9 ms; join/leave (§VIII-F)

between them, each of which we met as an incident in a running fleet. We state them as arrows (Fig. 1).

a) R1 needs R2 needs R4: An offer's price is a log-probability. The precondition of a pragmatic offer (is the door open?) is only meaningful if the posterior behind it is calibrated, and that posterior was only calibrated once the self-model's extrinsics were solved: a boresight error of 0.8° in one camera put every corner it saw a few centimetres off, in a way that averaged into a smaller declared uncertainty rather than a larger one. Actionable implies calibrated implies a self-model.

b) R2 needs R3: Epistemic self-construction with several writers requires ownership. Without it, two agents held different values of "the target" and no term in either state could reveal the disagreement; the return channel between them was a boolean. The cure was an edge the consumer owns plus an epoch, which is a statement about ownership, not about inference.

c) R3 needs R5: Ownership is only enforceable if the transport guarantees that a graceful exit runs cleanup and that

a transient peer flap does not delete a live node. A one-second flap at startup made an agent tear down its own nodes and exit cleanly; the fix was a debounced presence protocol, which is a transport property.

d) R5 needs R2: Bandwidth is bounded only because the graph carries beliefs, a mean and a covariance per concept, rather than sensor data. The media plane exists so that the graph does not have to.

e) R4 needs R1: Acting is the calibration experiment. There is deliberately no "calibrate" affordance; the self-model learns from the corrections that ordinary offers generate. A robot that never acts never calibrates.

III. RELATED WORK, BY REQUIREMENT

We organise prior work by which requirements it meets rather than by system, and summarise in Table II.

a) Scene graphs as records: The 3D Scene Graph [1] introduced the layered structure (building, rooms, objects, cameras); Kimera and its dynamic scene graphs [2], [7]

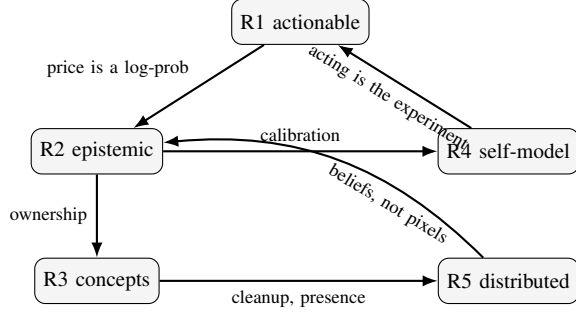


Fig. 1. The requirements as a dependency graph. An arrow $A \rightarrow B$ reads “ A needs B ”. Each arrow is backed by a measured incident (§II-A). The cycle is why the set cannot be satisfied one requirement at a time.

and Hydra [3], [8] built it online from a metric-semantic reconstruction; ConceptGraphs [4] and HOV-SG [9] made the labels open-vocabulary; Khronos [10] added time. All are single-pipeline systems in which the graph is an output. None exposes the uncertainty of a node as something the robot acts to reduce, and none carries actions in the graph.

b) *Rooms and walls in the loop*: S-Graphs [11], [12] is the closest work on R2 and R3 for the room layer: walls and rooms are factors in one optimisation, so the structure constrains the pose. It is still a single estimator and the graph is still passive.

c) *Task-driven graphs*: Clio [13] clusters the graph to the task at hand; SayPlan [14] plans over a scene graph with a language model. In both the task lives outside the representation and the graph is what the task reads. In an ASG the task is a prior added to the same functional the graph already minimises.

d) *Active SLAM and active inference in robotics*: The active-SLAM survey [15] frames exploration as decision-making under uncertainty and makes the point we lean on in §VIII: an open-loop dataset cannot score a closed-loop policy. Active inference [16], [17] supplies the one functional we use; its robotics applications [18], [19] have concentrated on control and manipulation rather than on the representation.

e) *Our own prior work*: CORTEX [20], [21] is the distributed working-memory substrate we inherit. The concept-first architecture of [6] showed rooms and doors as agents; the IWAI abstract [5] stated the actionable graph with a Fisher-information selector. Neither gave the requirements, the coupling, the pricing, or the self-model.

f) *The term*: “Active scene graph” is used in computer vision for active learning of image scene-graph generators. **[TODO: search and cite one representative; disambiguate in a footnote]**. We use the word in the control sense: a graph that acts.

IV. DEFINITION: THE ACTIVE SCENE GRAPH

A. Nodes are concepts

A node n is a tuple $(c, \mu_n, \Sigma_n, \Pi_n, \ell_n, a_n)$: a class c , a posterior mean and covariance over the class’s parameters, a precision Π_n attached to the observation channel that feeds it, a log-odds of existence ℓ_n , and the identity a_n of the agent

TABLE II
SYSTEMS AGAINST REQUIREMENTS. ● MET, ○ PARTIAL, BLANK NOT ADDRESSED. **[TODO: VERIFY EACH CELL AGAINST THE CITED PAPER BEFORE SUBMISSION.]**

System	R1	R2	R3	R4	R5
3D Scene Graph [1]					
Kimera, Hydra [2], [3]		○			○
S-Graphs [11]		○	○		
ConceptGraphs [4]					
Clio [13]	○				○
SayPlan [14]	○				
Concept-first rooms [6]		○	●		●
This work	●	●	●	●	●

that owns it. The parameters are class-specific: a room is a polygon of corners with a pose; a table is a box with a height; a bottle is a cylinder; a door is a hinge line, a width and an opening angle. Each class comes with a generative model of what the sensors return given the parameters, written as a signed distance field so that the likelihood is differentiable in the parameters: the room’s polygon SDF against LiDAR returns, a table as a smooth log-sum-exp box top with four cylindrical legs, a bottle as a cylinder. The same SDF serves for the fit and, projected through the camera model, for the expected information gain of a candidate view.

B. Edges are typed and timestamped

Geometry is carried by RT edges holding a rigid transform, a timestamp and a covariance, so that a consumer can extrapolate a lagging frame with the producer’s twist and can weight a transform by how much it is trusted. Three further edge types carry the active part: `has_intention` from a concept node to the affordance nodes it publishes; `executing` from the controller’s own node to the affordance it is currently executing; and `proto`, a self-edge marking a node whose identity is not yet resolved (a room the robot has just entered through a door and has not yet recognised as new or as one it has been in before).

C. Agents are owners

Every node is created, updated and removed by exactly one agent, and the agent’s identity is stored on the node. The agent is the process that runs the class’s generative model. It is a peer in a distributed graph, not a plugin in a pipeline: it starts, declares which peers it needs, serves while they are present, and cleans up its own nodes when it stops.

D. The lifecycle contract is the operational semantics

What an agent may do to the shared graph is fixed by a contract with five stages. *Create*: a node is born only from a tracker that has associated several frames, never from a single detection. *Update*: a frame may move geometry only if its frame is resolvable, its subject is centred in the image and the robot is still enough that the interaction-matrix variance does not dominate. *Remove*: the only removal evidence is “predicted visible but absent”, weighted by $P(\text{detect})$ and

requiring line of sight through the walls; occlusion holds the belief and a miss counter is forbidden. *History*: no agent may use a constant process noise; the volatility of a concept is inferred from its own history. *Ownership*: the identity stored on the node must match the agent that writes it. A conformance audit (§VIII-D) checks every agent against every stage.

V. ONE OBJECTIVE, TWO REGIMES

Every quantity the controller compares is a log-probability. There is one functional, the expected free energy of a policy π ,

$$G(\pi) = \underbrace{-\mathbb{E}_q[\log C(o_\pi)]}_{\text{pragmatic}} - \underbrace{\mathbb{E}_q[H(s) - H(s | o_\pi)]}_{\text{epistemic}}, \quad (1)$$

and the task layer differs from the representational layer only by the prior preference C .

With a flat C only the epistemic term survives. This is the fleet’s default: move so as to represent, and stay aligned with the world. It is the bottom, always-on layer, and it is what an ASG does when nobody has asked it for anything. With a sharp C over a goal state (“the robot is in room B”) the pragmatic term dominates, and an epistemic offer off the route is taken exactly when its information gain exceeds the pragmatic loss of the detour, in the same units. “Take the detour if it does not perturb the goal too much” is this inequality, not a rule.

A. The price table

Table III lists each term the controller sums, who produces it, and which hand-set constant it retires. The last column is the honest one: it names the constants that are still in the code and what should replace them.

Two properties make the table work across a fleet. Each producer computes ΔH from its own Σ , so the units are homogeneous by construction and no cross-class scale is needed. And the executor never learns what a door is: the affordance’s kind is a label used for logging, never a dispatch key, so a new concept adds offers without touching the controller.

B. Interruption is re-scoring

Selection, horizon and interruption are one comparison. The incumbent, if any, is the offer under an `executing` edge; the challenger is the minimiser of G over everything offered; a switch fires when the incumbent’s price exceeds the challenger’s by the commitment margin. Producers decay their epistemic gain as their Σ shrinks during execution, so an incumbent that is nearly resolved loses to a fresh challenger without a rule saying so; the interruption emerges from the value dynamics. The margin is the one constant left in this part of the system, and Table III says so.

VI. THE ROBOT AS A CONCEPT

The robot is a node in the graph with a frame tree beneath it: base, sensor mounts, and the rigid transforms between them. What distinguishes it from a table is only that its parameters are also the parameters of the measurement model of everything else. R4 says these parameters are beliefs, and that they are updated online.

a) *What is learnable*: On the platforms we use, low-level velocity loops hide the dynamics, so there is only a kinematic model to learn: wheel-odometry scale and rotation gain, and the extrinsics of each camera relative to the LiDAR. Dead reckoning cannot calibrate itself, because an odometry error and a true motion are indistinguishable from inside the odometry. What breaks the symmetry is an absolute reference, and the localiser already has one: the room polygon, against which the SDF anchors the posterior. Its error does not grow with distance travelled, so it is an independent witness of true motion.

b) *The free teacher*: The localiser’s own correction is the teacher. Every cycle it predicts the pose from the motion model and corrects it from the map; the correction is the residual the calibration must explain. A batch estimator over episodes solves jointly for the kinematic and extrinsic parameters that minimise the corrections, under the same free energy the localiser minimises over states. The parameters are separable only if each loads on a different component of the correction times a different covariate, which is why rotation gain (loads on heading, covariate angular speed) and translational scale (loads on position, covariate linear speed) can be told apart, and why the mount height of a camera cannot be separated from the height of the ceiling it sees (§VIII-E).

c) *Why there is no calibration affordance*: It would be natural to publish “calibrate yourself” as an offer. We do not, and the reason is the arrow R4→R1 in Fig. 1: the free energy of the self-model does not fall because the robot computed harder, it falls because the robot moved and the map disagreed. We measured this directly: across an injected 0 to 10% scale error the solver’s iteration count was flat at 13.3 to 13.9, while the correction load $|\text{estimated} - \text{predicted}|$ tracked the error. Acting is the experiment; the ordinary offers generate all the evidence the self-model needs.

d) *Downward coupling*: A correction to a parameter propagates into every belief that was fitted through it: the corrected extrinsics re-price every offer that depends on where a camera points. The rule is that this propagation may only loosen a belief, never tighten it; slower convergence is the correct behaviour. This is specified and not built (Table IV).

VII. ARCHITECTURE

R3 and R5 are met by an architecture with four parts.

a) *The graph*: A CRDT graph replicated on every agent’s machine, inherited from CORTEX [20], [21]. Each agent holds a full copy; writes are merged without a coordinator; signals fire on every peer when a node or edge changes. Every attribute is typed at compile time.

b) *The media plane*: Sensor streams do not enter the graph. A producer advertises, on the sensor’s node, a descriptor naming a topic on a dedicated DDS domain; consumers subscribe to it zero-copy. The graph carries beliefs and the descriptor; the plane carries pixels and points. This is the R5→R2 arrow: bandwidth is bounded because the representation is parametric.

c) *The presence protocol*: An agent declares the peers it requires, starts serving only when they are present, and

TABLE III
EVERY TERM THE CONTROLLER COMPARES, IN NATS. THE LAST COLUMN NAMES THE CONSTANT EACH TERM RETIRES, OR THE CONSTANT THAT REMAINS.

Term	Price	Producer	Retires / remains
Epistemic value of an offer	$P(\text{detect}) \cdot \Delta H$, with $\Delta H = H(\Sigma) - H(\Sigma^*)$ under the concept's own model	each concept, from its own Σ	per-class gain scales. One survives (<code>aff_room_gain_scale=0.35</code>); if a scale is needed the units are not homogenised, so it must go.
Precondition	$\log p(\text{precondition})$; a door open with $p = 0.3$ costs -1.2 nats; $p = 0$ is not an offer	pragmatic-affordance producer	on/off gating (presence of a precondition rather than its probability)
Navigation cost	expected growth of self-entropy along the path, $\frac{1}{2} \log \det(\Sigma_{\text{pose}} + Q(\text{path})) - \frac{1}{2} \log \det \Sigma_{\text{pose}}$, with Q from the self-model's calibrated motion noise (R4), plus $\log(1 - p_{\text{coll}})$ along the path, plus delay against C	controller, from the robot concept	$\lambda \cdot d_{\text{nav}}$ with $\lambda = 0.2$ nats/m: a constant that should be a function of the calibrated Q . Still in the code; kept as an A/B flag.
Commitment	a hysteresis margin: switch only if $G(\text{incumbent}) - G(\text{challenger}) > m$	controller	$m = 0.5$ nats remains. Derivable as the expected loss of the half-executed policy; not done.
Existence	log-odds; absence weighted by $P(\text{detect})$; occlusion adds 0	shared existence module	miss counters, debounce timers
Publication gate	the adequacy gap between the posterior Σ and Σ^* , in nats	each concept	tuned publish bounds

degrades when one leaves. Degradation is debounced: a transient flap does not trigger cleanup. The protocol was modelled in TLA+ and checked exhaustively: eight safety invariants and four liveness properties over 65 536 states for a three-agent instance. The model abstracts heartbeat timing away, and the failure that later stopped the fleet was a timing failure (the undebounded flap). A proof about a specification and a specification that survives contact are different achievements, and we report both.

d) *The affordance protocol*: A producer publishes an offer with a target and an epoch; the controller claims it by writing an `executing` edge it owns, carrying the epoch and the pose it is actually driving to. The producer measures progress against the consumer's stated target, never its own latest proposal, and a mismatch of epochs is detectable by both sides. The earlier version of this protocol returned a boolean, and the TLA+ model of it reproduces the failure in a four-state counterexample; with the claim carried, the model is clean at three times the bound (22 815 states). The invariant that matters is general: no party may judge another by a quantity the other cannot observe.

VIII. EVALUATION

A. Premise

A dataset comparison does not apply to this work, and we say so rather than supply one. A dataset is an open-loop recording; under (1) the observations received depend on the actions chosen, so replaying a fixed sequence scores one policy on data another policy generated. The comparison that is meaningful is policy against policy inside one closed loop, with body, sensors, noise, timing and environment held fixed and only the decision rule changed [15]. Our comparators are the heuristics each mechanism replaced: each was written by

people who wanted it to work, tuned until it did, and shipped, so it cannot be called a weakened baseline.

Platforms: a Webots simulation of a two-room apartment running the same component code as the robots; a differential-drive robot (Shadow) with a 360° LiDAR, a stereo camera and a panoramic camera; and a second differential-drive robot (P3Bot) with a different LiDAR and no panoramic camera. The fleet is the same on all three; the robot concept differs.

B. R2: the uncertainty is consumed

a) *Calibrated corners on a physical robot*: The room concept declares a corner publishable when its posterior σ falls under a bound. On the physical robot, against a surveyed room, a corner declared publishable has a median true error of 2.0 cm and lies within its declared 6 cm in 86 % of cases; a withheld corner has 3.6 cm and 64 %. The gate is doing what the requirement says: the declared uncertainty separates good corners from bad ones. The full study is in a companion paper; we do not repeat its figures.

b) *Room shape from an epistemic explorer*: On 50 synthetic rooms with adversarial spurs and alcoves, replacing a coverage heuristic by the information-gain explorer raised the median layout IoU from 0.947 to 0.954 and the worst room (an L shape) from 0.812 to 0.944.

c) *The null, and why*: On 594 real annotated floor plans, paired by room, the same explorer against the same heuristic gives a mean IoU difference of -0.0007 with a 95% confidence interval of $[-0.0019, +0.0004]$; the heuristic wins 263 rooms, the explorer 251, with 69 ties. The epistemic term buys nothing on this corpus. The cause is the endpoint: every room in both arms ran to the 900-frame budget (594 of 594 and 593 of 594), so the final map was measured at saturation, where any reasonable policy has already seen every wall. An objective cannot be graded on an endpoint its own budget has

TABLE IV
LEDGER: WHAT IS DESIGNED (D), IMPLEMENTED (I) AND EVALUATED (E).

Mechanism	D	I	E
Concept nodes with (μ, Σ, Π, ℓ) , SDF likelihoods	•	•	•
Epistemic offers priced by $P(\text{detect}) \cdot \Delta H$	•	•	•
Pragmatic offers (door approach / open / cross)	•	•	
$\log p(\text{precondition})$ instead of on/off	•		
Task prior C in the controller	•		
Navigation cost from calibrated Q (replacing λ)	•		
Existence as $P(\text{detect})$ -weighted log-odds	•	•	• ^a
Room structural adoption judge	•	•	•
Lifecycle contract and conformance audit	•	•	•
Self-model: wheel covariance, preintegration	•	•	•
Self-model: closed-loop camera extrinsics	•	•	•
Self-model: batch estimator over episodes	•	•	•
Downward coupling (loosen-only propagation)	•		
Proto-room: navigate while the layout is estimated	•	• ^b	
Presence protocol (TLA+)	•	•	•
Affordance claim protocol (TLA+)	•	•	•
Media plane, zero-copy	•	•	•
Bounded CRDT history			

^a evaluated for tables, doors, cabinets, refrigerators; not yet for bottles.

^b built and compiled; first live test pending.

saturated. The number that predicts the null is the 594/594 at the cap, and it was available before the run. What is supported is that the uncertainty is consumed: it gates publication and the explorer reads it. What is not supported is that the epistemic term buys accuracy or speed over a simple heuristic at single-room scale. Tighter budgets and multi-room environments are where a difference would have to appear, and they are untested.

C. R1: one executor, offers from any concept

The affordance claim protocol is evaluated formally (§VII) and live: after the epoch and the owned edge were introduced, the refusal loop in which a re-armed offer looked new to the controller every cycle fell from 47% of decisions to 7%, and the protocol-event flood the producers generated to reconcile their view of the target (12 per second) went to zero. The door offers (approach, open, cross) are built and compile against the same executor with no door-specific code in the controller; they have not yet been run live, and Table IV says so. The task-level experiment the outline calls for, one room, one door, one live epistemic offer, with the controller logging the decomposition of G per candidate per decision, is the first item of future work.

D. R3: the lifecycle contract has teeth

A conformance audit checks each of eight concept agents against the five stages of the contract and three sub-conditions (admissibility of a frame, $P(\text{detect})$ -weighted absence, line of sight through walls). At the audit of 2026-07-31, five agents conformed on every column, two conformed with an attributable history missing, and one (the human concept, inherited) failed every stage but ownership; it is scheduled for rewrite rather than reported as conforming. The clearest effect of the contract is on phantoms: bringing the bottle concept under the create stage (birth from a tracker, never a single frame) reduced phantom births on a fixed tour from 130 to 1. Ownership is enforced at startup: an agent refuses to start if the node it would own carries another agent's identity.

E. R4: the self-model learns from its own corrections

a) *Extrinsic attribution:* On one recorded tour we inject a 1° yaw error into the panoramic camera, into the stereo camera, and into the LiDAR in turn, and read what the closed-loop extrinsic estimator recovers. The stereo camera's yaw is recovered at 0.984 ± 0.018 of the injected value, the panoramic camera's at 0.454 ± 0.175 , and an error injected into the LiDAR shows up in both cameras and not in their closure (-0.018), which says which device is misaligned. The asymmetry between the cameras is itself a result: on a panorama a yaw is a constant pixel shift, indistinguishable from a corner's own offset, while on a pinhole the shift grows towards the field edge as $\sec^2 \theta$, so one corner seen across the image separates yaw from offset by itself. The panorama's uniformity, which is what lets it separate pitch from height, is exactly what costs it yaw. The pair is complementary.

b) *A bias the precision could not see:* The estimator found a boresight error of $+0.814^\circ$ in the stereo camera that the corner channel had been absorbing as a smaller declared σ rather than a larger error. Applying it is what let the corner calibration above hold. This is the $R1 \rightarrow R2 \rightarrow R4$ chain measured.

c) *What must not be published:* The same estimator produces a mount height and a pitch for each camera. We do not publish them. Floor corners and ceiling corners disagree by 116 mm about the height of one mount, while agreeing on its yaw to 0.0006° ; the height channel is measuring the ceiling constant of the room, not the mount. A parameter is only learnable if it loads on a covariate nothing else loads on, and here it does not.

d) *Odometry:* Wheel-odometry covariance is injected per cycle as a function of speed, and the rotational variance had to be raised by a factor of 580 from the vendor's constant before the localiser's normalised innovation was consistent. A zero-velocity update whose σ was measured on one robot proved 5.9 to 7.3 times too tight on the other, which is the reason the robot concept, not a shared constant, owns it.

F. R5: latency and membership

Table V gives the age of each stream at the consumer, before and after two changes to the producers: a drift-free

TABLE V
STREAM AGE AT THE CONSUMER, MS, MEASURED ON 2026-09-10.

Stream	before	after
LiDAR (Helios)	35.2	8.8
LiDAR (Bpearl)	–	10.0
Panoramic RGB	50.2	11.1
Stereo RGB-D	36.0	9.0

trigger aligned to the sensor’s acquisition grid, and duplicate suppression. The floor is not the transport: the data is already 17 ms (LiDAR) to 27 ms (RGB-D) old when the simulator hands it over, and the plane adds under 10 ms above that. The LiDAR stream runs at 20 Hz, its source rate.

Membership: agents are started and stopped in arbitrary order in every session; a graceful stop removes the agent’s owned nodes and a restart re-adopts orphaned ones under the ownership rule. The defect we know of is stated in §IX.

G. Localisation and object beliefs against ground truth

Two numbers from the same fleet, reported so that the beliefs the graph carries can be judged as beliefs. Against simulator ground truth over 4 runs and 5468 poses, the room-anchored localiser has a median position error of 72 mm, 145 mm at the 90th percentile, and 0.8° in heading, at 19.1 Hz. For the bottle concept, against ground truth over 2173 rows, the position RMSE is 13.6 mm with a median normalised estimation error squared of 0.46: the covariances are honest, which is what a belief-based graph must show before its offers can be priced.

IX. LIMITATIONS AND CLAIM DISCIPLINE

- The epistemic term is not shown to beat a heuristic at single-room scale, and we have explained why the endpoint could not show it. The multi-room, budget-limited experiment is not done.
- The task-level functional (a sharp C) is designed and not implemented; the R1 evaluation is of the protocol, not of a task.
- Two constants remain in the controller: the commitment margin, and λ until it is replaced by the calibrated Q . Both are named in Table III.
- The CRDT history is unbounded: the per-node version vector grows with every write, and on long sessions the merge cost was measured at one full core on 318 000 entries. Either it is bounded or the claim is scoped to session length; today it is the latter.
- The detectability envelope $P(\text{detect})$ is fitted in simulation for the classes reported here; its calibration on real images is reserved for a separate paper and not claimed.
- The self-model’s translational scale inherits the map’s error: a room polygon 2% too large hands 2% to the scale, confidently. Rotation is immune; translation is only as good as the survey.
- Doors: the proto-room mechanism that lets the robot navigate while a new room is being estimated is built and compiled but has not been run live.

X. CONCLUSION

An Active Scene Graph is a scene graph whose nodes are beliefs, whose beliefs publish what would reduce them, whose robot is one of its own beliefs, and whose controller prices everything in one currency. We have argued that the five requirements this entails are coupled, shown a system that meets them together, and reported what its evaluation supports and what it does not. The next version of this preprint adds the task-layer experiment on the physical robot and the multi-room setting in which an epistemic explorer could be expected to matter.

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